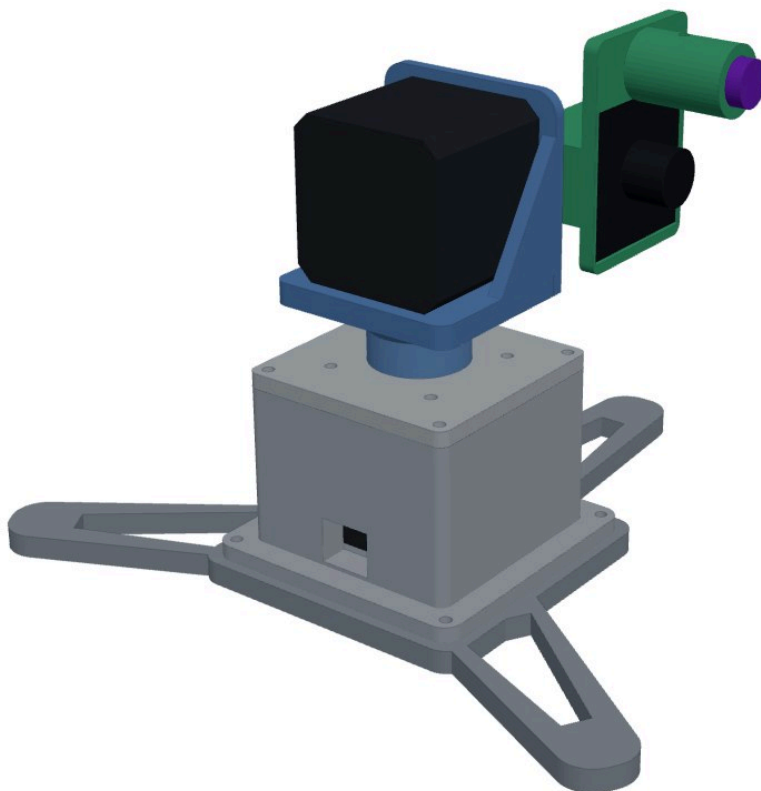


Two big stepper motors. One tripod. Nothing to solder.

The direct-drive pan-tilt tower, built from the design you chose: the pan motor stands inside a printed tower, the tilt motor lies on top of it, and the camera and laser clamp straight onto the tilt shaft. A CNC shield plugs into an Arduino UNO, so there is no breadboard.

**0.11°**

smallest move per motor
step (1/16 step)

198 g

to print, six parts, PETG

19

clearance checks, 0
collisions (pan $\pm 90^\circ$, tilt
 $\pm 45^\circ$)

₹2,260–4,050

all in, parts + print shop
(estimate)

Honest status. This design is checked on the computer only. It is not printed or tested yet. Print the test coupon first, then the parts. Prices are estimates: confirm them at the shop.

We already own most of it

Already have ✓

2 × A4988 stepper drivers

from MK2. Test each one first (the self-test does it).

12 V adapter

check the label: 3 A or more is best. A 2 A adapter works if you set the driver current low. Replace any damaged cable.

2 × NEMA17 motors + their 4-pin cables

from MK2. The shaft must be 5 mm.

USB camera (38 × 38 mm board)

from MK2. Measure your board: the pocket is 38.6 mm.

Beacon + decoy

no change.

Laptop with the Live page

no change.

Not needed in this design

No bearings, belts or pulleys. No angle sensors, no splitter.
No breadboard, no soldering. No metal shaft hubs (the hubs are printed and clamp with a set screw).

Why it is cheaper than the belt design

The belt design needs 4 bearings, 3 pulleys, 4 belts and 1,074 g of printing. The tower needs none of those parts and prints 198 g.

Buy list

ITEM	QTY	PRICE (ESTIMATE)	WHERE	NOTE
☐ Arduino UNO R3 + USB-B cable	1	₹450–700	Amar shop	CH340 clone is fine. Amazon: "Arduino UNO R3 CH340 with cable".
☐ CNC Shield V3 for UNO	1	₹250–350	Amar shop	It must say "V3" and fit on an UNO. Amazon: "CNC shield V3 Arduino UNO".
☐ 12 mm laser module, red 650 nm, 5 mW, 5 V	1	₹100–200	Amar shop	Only 5 mW. Never a "burning" laser. Amazon: "650nm 5mW laser module 12mm".
☐ Dupont jumper wires, female-female, 20 cm	1	₹60–100	Amar shop	One pack. For the laser.
☐ Fuse holder 5×20 mm + 3 A fuses + rocker switch	1	₹60–120	Amar shop	Safety for the 12 V. Optional but cheap.
☐ Heatsinks for A4988 (if the drivers have none)	1	₹20–40	Amar shop	Small stick-on type.
☐ M3 screw kit (6 to 30 mm) with nuts and washers	1	₹250–400	Hardware shop	Covers every screw below. Amazon: "M3 screw nut assortment kit".
☐ M2 × 6 self-tapping screws	1	₹20–40	Hardware shop	Four are used for the camera board.
☐ Super glue + rubber feet (or non-slip tape)	1	₹60–120	Hardware shop	Three feet under the tripod legs.

If a motor or driver from MK2 turns out to be damaged: NEMA17 about ₹400–600 each, A4988 about ₹150–200 each. The shop slips for carrying are in a separate short PDF.

Go shopping: Amar shop first, Amazon as backup

Take the separate **shop sheet** (3 pages) with you. It has one slip for each place. Quick version:

WHERE	WHAT TO GET	ABOUT
Amar shop electronics	UNO R3 + cable · CNC shield V3 · 12 mm laser module · jumper wires · fuse holder + fuses + switch · heatsinks	₹940–1,510
Hardware shop	M3 screw kit with nuts · M2 × 6 self-tapping · super glue · rubber feet	₹330–560
3D print shop	The six parts (+ the coupon first), PETG, 198 g	₹990–1,980
Amazon only if needed	Anything the shops do not have. Search words are on shop slip 2.	same

Budget

PART OF THE PROJECT	COST (ESTIMATE)
Parts to buy (list above)	₹1,270–2,070
Printing 198 g at a print shop (₹5–10 per gram)	₹990–1,980
Test coupon (28 g), print first	₹140–281
Total with a print shop	₹2,400–4,331
Total if a friend prints it (PETG only)	₹1,510–2,470

The order that saves time

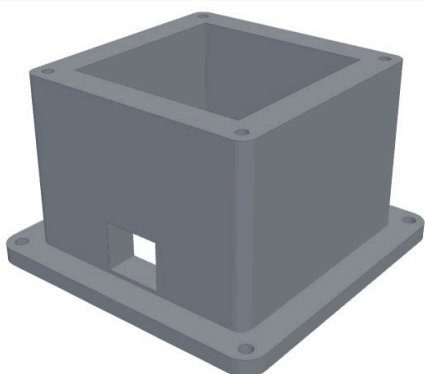
1. **Today:** send the coupon STL to the print shop. It is small and quick.
2. **Same day:** buy the electronics and the screw kit.
3. **When the coupon fits:** print the six parts. The parts take about 14–20 hours.
4. Wire the UNO and CNC shield while the printer runs, and test each motor on the bench.

Safety. The laser is 5 mW, red. Never look into the beam and never point it at a person or at a shiny surface. The 12 V adapter needs an undamaged cable and a fuse. Never plug or unplug a motor or a driver with 12 V on.

What to print, and how

PART	QTY	SIZE MM	WEIGHT	SUPPORTS	HOW TO LAY IT
Tripod base 01_tripod_base.stl	1	190 × 167 × 6	52 g	No	Flat on the bed, as shown. Three legs and the square pad. Four nut pockets on the underside.
Tower sleeve 02_tower_sleeve.stl	1	74 × 74 × 46	66 g	No	Foot flange on the bed, open end up. Holds the pan motor. A cable slot near the bottom.
Tower cap 03_tower_cap.stl	1	60 × 60 × 4	16 g	No	Flat on the bed. The pan motor screws to its underside.
Pan hub 04_pan_hub.stl	1	34 × 34 × 16	9 g	No	Small end down, wide end up. Clamps on the pan shaft. Slot for one M3 nut.
Tilt mount 05_tilt_mount.stl	1	52 × 47 × 51	26 g	No	Plate down, wall and ribs standing up. The tilt motor screws to its wall.
Head 06_head.stl	1	58 × 40 × 70	29 g	Yes, only under the camera plate	Stand it on the rear of the clamp, laser tube pointing up. Camera on the front, laser tube on top, clamps on the tilt shaft.
Test coupon (print FIRST) 00_test_coupon.stl	1	100 × 46 × 8	28 g	No	Flat on the bed. Checks every fit in about an hour. See page "Do the parts fit?".

Settings: Material PETG (PLA is fine for the first test). Layer 0.2 mm. 4 walls. Infill 25%. No brim needed.
Total: 198 g for the six parts. Weights are estimates at 25% infill.



Left: tower sleeve. Right: tilt mount. The other parts are pictured on the shop sheet and in the exploded view.

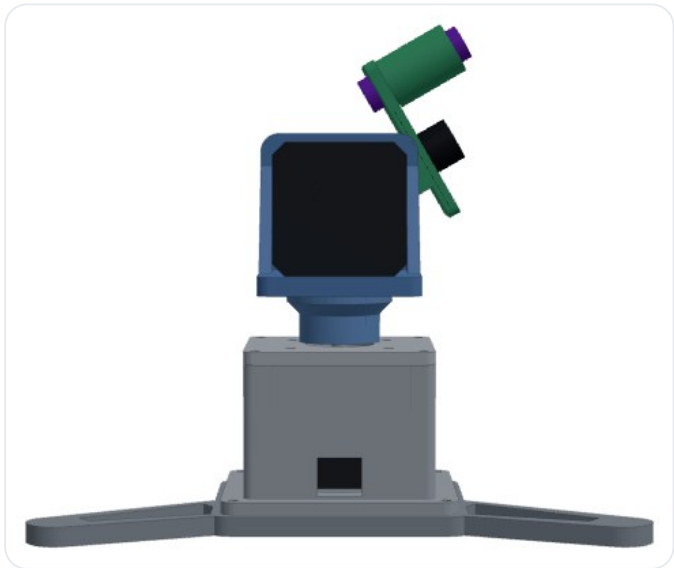
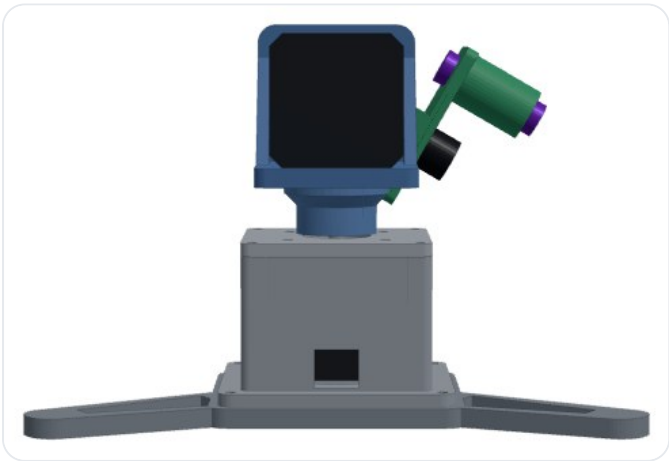
Why the head needs supports. The camera plate is wider than the arm behind it. Use tree supports and only under that plate. Do not support the bores; drill them out with a 5 mm and a 12 mm drill if they come out tight.

Every bought part against its printed hole

BOUGHT PART	ITS SIZE	OUR HOLE OR POCKET	GAP
NEMA17 motor body (pan)	42.3 mm square	Tower cavity 44.6 mm square	1.15 mm each side
NEMA17 pilot boss (pan and tilt)	22 mm	Hole in the cap and in the wall: 22.3 mm	0.15 mm each side
NEMA17 screw holes	31 mm square	4 clearance holes, 3.2 mm	M3 passes
Motor shaft (pan and tilt)	5 mm	Bore in the hub and in the head: 5.2 mm	0.1 mm each side
M3 nut (across flats 5.5)	2.4 mm thick	Slot 5.9 × 2.6 mm	drops in
M3 screw cutting its own thread	3 mm	Pilot hole 2.7 mm	bites
USB camera board	38 mm square	Pocket 38.6 mm square, 2.4 mm deep	0.3 mm each side
Laser module	12 mm	Bore 12.3 mm + one M3 clamp screw	push fit
Tower flange to tripod base	4 × M3 at 66 mm square	4 holes + 4 nut pockets underneath	bolts

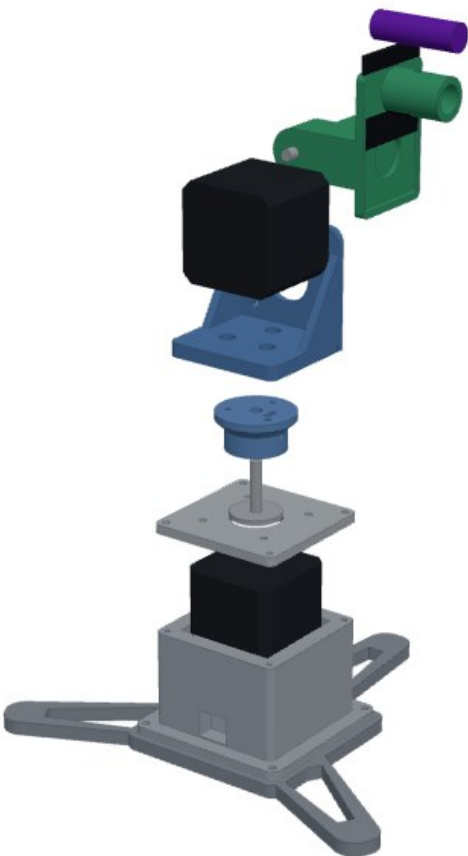
Moving clearance (computer check of the real 3D shapes)

CHECK	POSES	COLLISIONS
Standing still, every part against every part	1	0
Head and plate turning against the tower and base: pan -90° to +90°	6	0
Head against the tilt mount and plate: tilt -45° to +45°	6	0
Head against the tower and base: tilt -45° to +45°	6	0



Tilt -30° (left) and +30° (right). The head always stays about 2 mm away from the tilt wall, and the firmware stops at ±30°.

Exploded view



NO .	PART	HOLDS
01	Tripod base	Everything. Three legs, one pointing to the head side.
02	Tower sleeve	The pan motor hangs inside it.
03	Tower cap	The pan motor is screwed to its underside.
04	Pan hub	Clamps on the pan shaft and carries the tilt mount.
05	Tilt mount	The tilt motor is screwed to its wall, lying on its side.
06	Head	Clamps on the tilt shaft. Camera in front, laser on top.

Size when built: 190 × 167 mm footprint, 137 mm tall. The tilt axis is 100 mm above the table. The pan axis passes through the centre of the tilt motor, so the tower stays balanced.

Mechanics

- 1 **Print and check the coupon.** Print the test coupon. A NEMA17 shaft must slide into the 5.2 mm hole, an M3 screw must bite in the small hole, an M3 nut must drop into the slot, the motor pilot must sit in the 22 mm hole, the camera board must lie in the pocket and the laser must push into the round hole.

CHECK: All six fits are right. If one is not, tell us which, and we change that one number.

- 2 **Print the six parts.** Use the print job sheet. 198 g in total.

CHECK: Every part is flat where it should be and the holes are clean.

- 3 **Pan motor onto the cap.** Put the motor flange against the underside of the cap. The shaft goes up through the round hole. Fit 4 × M3 × 8 from the top.

CHECK: The shaft sticks out about 20 mm above the cap.

- 4 **Cap and motor onto the sleeve.** Feed the motor cable out through the slot. Lower the motor into the sleeve. Fit 4 × M3 × 12 into the four corners (they cut their own thread; do not over-tighten).

CHECK: The cap lies flat, with no gap.

- 5 **Sleeve onto the base.** Push 4 nuts into the pockets under the base. Stand the sleeve on the pad. Fit 4 × M3 × 12 from the top.

CHECK: The tower stands straight. The cable slot faces away from the head side.



Inside view: the pan motor hangs from the cap, the tower sleeve is hidden.

Top half and finishing

- 6 Pan hub.** Drop a nut into the hub slot. Slide the hub onto the shaft, wide end up. Turn it so the set screw meets the flat side of the shaft. Tighten $1 \times M3 \times 8$. Leave 1 mm above the cap.

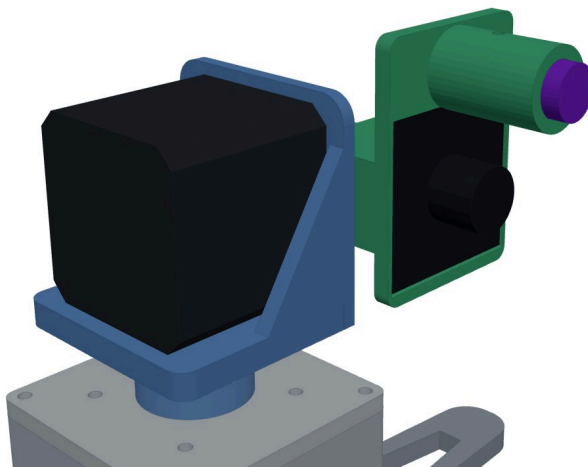
CHECK: Turn the hub by hand: the shaft turns with it. It does not rub the cap.
- 7 Tilt mount on the hub.** Put $3 \times M3 \times 10$ into the plate (the heads sit in the round pockets on top). Screw them down into the hub. The shaft tip goes into the small pocket under the plate.

CHECK: The plate is level and turns with the shaft.
- 8 Tilt motor.** Slide the motor in from the back, flat on the plate, its round boss into the hole in the wall. Fit $4 \times M3 \times 8$ from the front of the wall. The cable points backwards.

CHECK: The shaft sticks out about 20 mm beyond the wall.
- 9 Head, camera and laser.** Drop a nut into the head slot. Slide the head on the tilt shaft until it is 1–2 mm from the wall. Tighten $1 \times M3 \times 8$ on the flat. Mark and pre-drill the camera board corners, then fix it with $4 \times M2 \times 6$ (or a little glue). Push in the laser and tighten the top clamp screw.

CHECK: With the power off, the head swings by hand about $\pm 45^\circ$ without touching anything.
- 10 Wire it, flash it, test it.** Follow the next two pages: plug-in wiring, then the firmware and the tests.

CHECK: The Live page says the rig is connected.

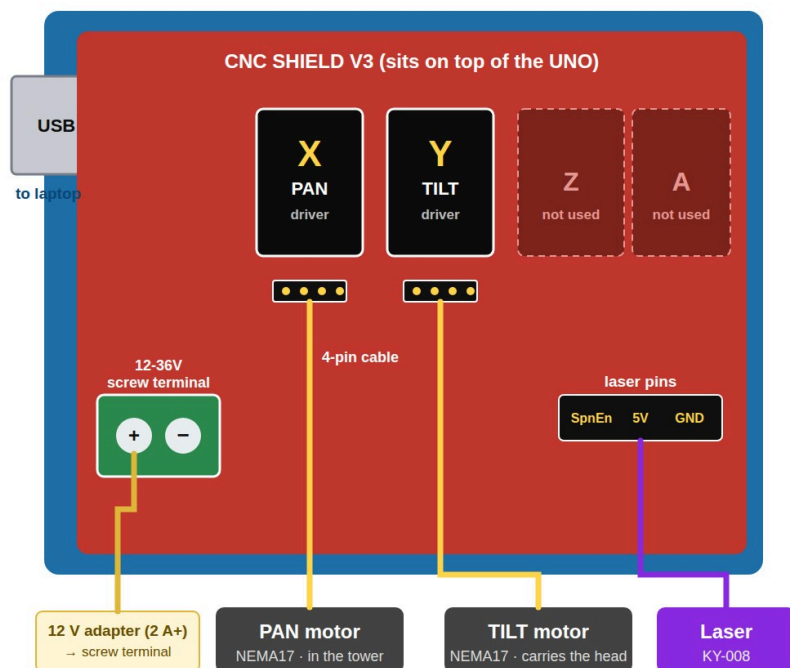


Head, camera and laser, close up.

Plug it in

Plug it in. No breadboard. No soldering.

Arduino UNO + CNC Shield V3 + 2 stepper drivers. Schematic, not to scale: always match the labels printed on your shield.



6 steps, about 20 minutes

- 1 Stack.** Press the shield onto the UNO.
- 2 Drivers.** Push two A4988 drivers into X and Y. Match the pin labels. Fit all 3 small jumpers under each (1/16 step). Stick on heatsinks.
- 3 Motors.** Pan motor into X, tilt into Y. Wrong way round? Flip that plug.
- 4 Laser.** Signal to SpnEn, – to GND. (jumper wires, no soldering)
- 5 Power.** 12 V adapter to the screw terminal. Check + and – first.
- 6 USB.** UNO to laptop. Flash the rig firmware, open the Live page, press CONNECT RIG.

Never do this with 12 V on:

- plug or unplug a motor or a driver
- put a driver in backwards (labels must match)
- power the motors from USB alone (USB = brain only)

Put the firmware on the UNO

1. Plug the UNO into the laptop with the USB cable. Power off the 12 V.
2. Run **tools/rig/flash.sh tower**. It builds the direct-drive version (ratio 1:1, pan $\pm 90^\circ$, tilt $\pm 30^\circ$, gentle speed).
3. The board prints **"# ZeroDrift Mk2 ready, encoders=no, belts=1:1"**. That line means it worked.

Set the Live page

1. Rig type: **MK2 · steppers**. Drive: **Direct drive (1:1)**.
2. Press **CONNECT RIG**, then **START CAMERA**. Turn on the 12 V.
3. Press **TEST MOTION**: pan and tilt must move the way the picture on screen moves. If one goes the wrong way, flip the motor plug, or use INVERT.
4. Switch on the laser and press **SET AIM** on the laser dot.

Driver current. Fit all 3 jumpers under each A4988 (1/16 step). Turn the small screw on each driver until the motor is strong enough not to skip steps. Keep the driver only warm. If a motor is too hot to touch, turn it down.

Is it working?

These are our targets. We have not measured them yet. Write the real result next to each one.

TEST	TARGET	RESULT
☐ Pan and tilt directions on TEST MOTION	Both right	
☐ Laser dot after SET AIM, beacon 2 m away	Dot on the beacon	
☐ Time to lock on the blinking beacon	About 2 s or less	
☐ Decoy next to the beacon, 10 tries	0 wrong locks	
☐ Cover the beacon, then uncover it	Holds, then locks again	
☐ Move the beacon left, right, up, down	The dot follows	
☐ 10 minutes running, then press C (centre)	Back to the same spot	

If something goes wrong

WHAT YOU SEE	WHAT TO DO
A motor buzzes and does not turn	Check the 4-pin plug sits fully in. Check the 12 V is on. Turn the driver current up a little.
A motor turns the wrong way	Flip that motor's plug, or use INVERT on the Live page.
A hub or the head slips on its shaft	The set screw must press on the flat side. Tighten it. Add a drop of thread-lock.
The head drops or shakes	The camera and laser are too heavy or off-centre. Keep the head under about 150 g.
It loses position (centre is not centre)	Turn the driver current up a little or lower the speed. Check the jumpers (1/16).
A driver gets too hot	Fit the heatsink, turn the current down, add a small fan.

If you need finer aim later

One step is 0.11° , about 2 pixels on the MK2 camera. For 0.056° , use DRV8825 drivers at 1/32 (set MICROSTEPS to 32 in the firmware). For 0.028° use the belt design, Plan Urena. The UNO, shield, motors and firmware carry over.